

// Arraylist

```
import java.util.ArrayList;

public class ArrayListCRUD {
    public static void main(String[] args) {

        ArrayList<String> list = new ArrayList<>();

        // Create
        list.add("Ahmed");
        list.add("Ali");
        list.add("Sara");

        System.out.println(list);

        // Update
        list.set(1, "Omar");

        // Delete
        list.remove("Sara");

        System.out.println(list);
    }
}
```

// LinkedList

```
import java.util.LinkedList;

public class LinkedListCRUD {

    public static void main(String[] args) {

        LinkedList<String> list = new LinkedList<>();

        // Add
        list.add("Ahmed");
        list.add("Ali");

        System.out.println(list);

        // Update
        list.set(0, "Mohamed");

        // Delete
        list.remove("Ali");

        System.out.println(list);
    }
}
```

// PriorityQueue

import java.util.PriorityQueue;

public class PriorityQueueCRUD {

public static void main(String[] args) {

PriorityQueue<Integer> pq = new PriorityQueue<>();

// Create

pq.add(30);

pq.add(10);

pq.add(20);

System.out.println(pq);

// Update (remove then add)

pq.remove(20);

pq.add(25);

// Delete

pq.remove(10);

System.out.println(pq);

}

}

//ArrayDeque

import java.util.ArrayDeque;

public class ArrayDequeCRUD {

public static void main(String[] args) {

ArrayDeque<String> deque = new ArrayDeque<>();

// Create

deque.add("A");

deque.add("B");

deque.add("C");

System.out.println(deque);

// Update (remove then add)

deque.remove("B");

deque.add("D");

// Delete

deque.remove("A");

System.out.println(deque);

}

}

//HashSet

import java.util.HashSet;

public class HashSetCRUD {

public static void main(String[] args) {

HashSet<String> set = new HashSet<>();

// Create

set.add("Ahmed");

set.add("Ali");

System.out.println(set);

// Update

set.remove("Ali");

set.add("Omar");

// Delete

set.remove("Ahmed");

System.out.println(set);

}

}

//LinkedHashSet

```
import java.util.LinkedHashSet;
```

```
public class LinkedHashSetCRUD {
```

```
    public static void main(String[] args) {
```

```
        LinkedHashSet<String> set = new  
        LinkedHashSet<>();
```

```
        // Create
```

```
        set.add("Ahmed");
```

```
        set.add("Ali");
```

```
        System.out.println(set);
```

```
        // Update
```

```
        set.remove("Ali");
```

```
        set.add("Omar");
```

```
        // Delete
```

```
        set.remove("Ahmed");
```

```
        System.out.println(set);
```

```
    }
```

```
}
```

//HashMap

```
import java.util.HashMap;

public class HashMapCRUD {

    public static void main(String[] args) {

        HashMap<Integer, String> map = new HashMap<>();

        // Create
        map.put(1, "Ahmed");
        map.put(2, "Ali");

        System.out.println(map);

        // Update
        map.put(2, "Omar");

        // Delete
        map.remove(1);

        System.out.println(map);
    }
}
```

//Stack

```
import java.util.Stack;
```

```
public class StackCRUD {
```

```
    public static void main(String[] args) {
```

```
        Stack<String> stack = new Stack<>();
```

```
        // Create
```

```
        stack.push("Ahmed");
```

```
        stack.push("Ali");
```

```
        stack.push("Sara");
```

```
        System.out.println("Stack: " + stack);
```

```
        System.out.println("Top Element: " + stack.peek());
```

```
        // Update
```

```
        int index = stack.indexOf("Ali");
```

```
        if (index != -1 ) {
```

```
            stack.set(index, "Omar");
```

```
        }
```

```
        // Delete
```

```
        stack.pop(); Here will remove Sara
```

```
        System.out.println("Updated Stack: " + stack);
```

```
    }
```

```
}
```